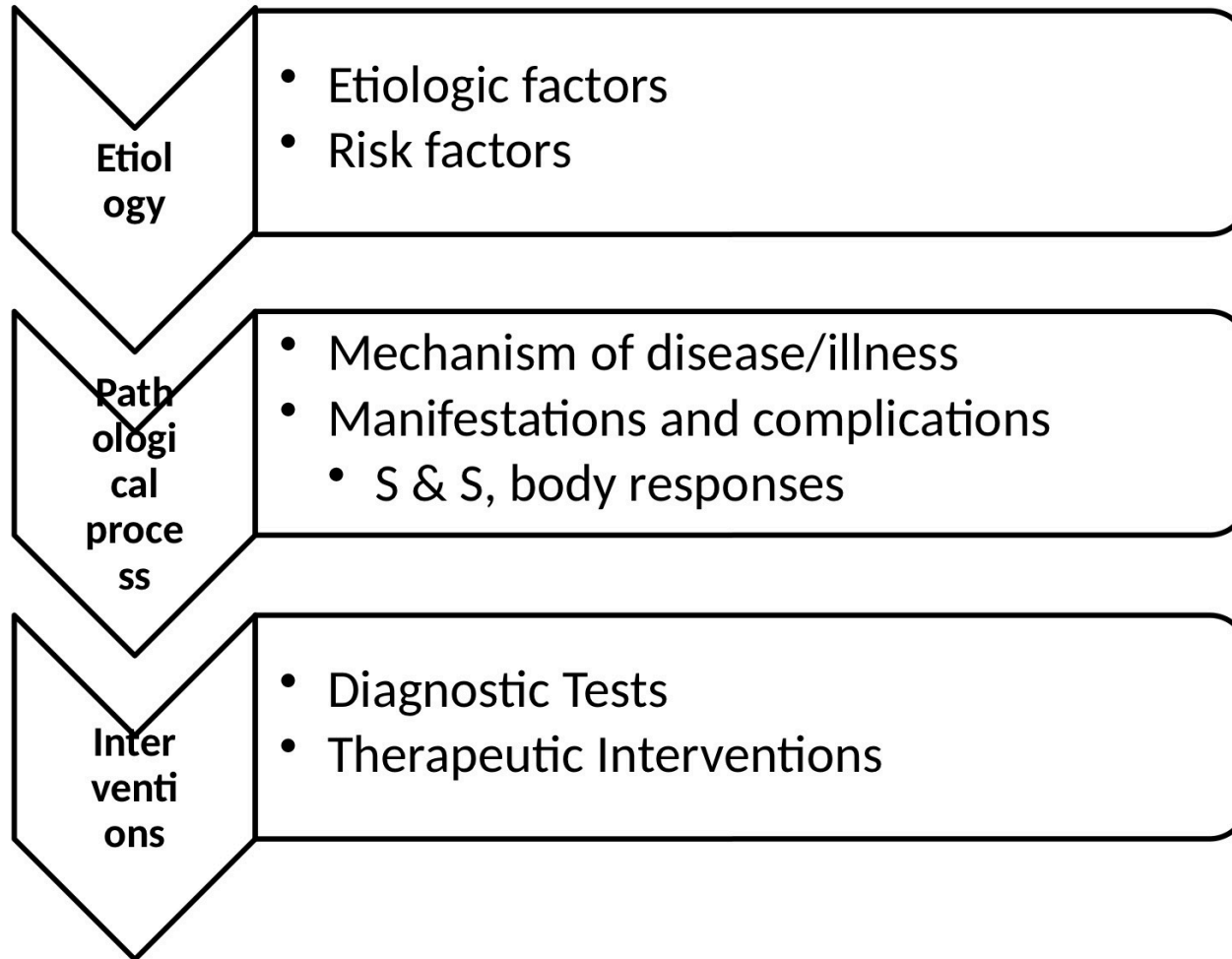


Pathology & Therapeutics II:

Review and Shock

Week 1

Pathophysiology Framework- Review



- Do you remember what these terms mean?
- Can you give an example for each?

Review

- What is cardiac output? (CO)
- What is mean arterial pressure? (MAP)
- What is systemic/peripheral vascular resistance (SVR/PVR)?

Review

- Do you remember the autonomic nervous system?
 - What is released when the sympathetic nervous system is activated?
- Do you remember the renin-angiotensin-aldosterone system (RAAS)?
 - What is the end result?

Circulatory Failure (AKA SHOCK)

How

- * cardiac pump
- * Vascular
- * Adequate blood volume

When $CO = HR \times SV$ is elevated
 ↑ ↓

It is trying to compensate
to maintain the "CO" normal

What is CO/SV?

-
-

What is MAP? (Mean Arterial Pressure)

-

What is SVR/PVR?

-

Circulatory failure (Shock)

- A condition in which **tissue perfusion** is inadequate to deliver **oxygen and nutrients** to support vital organs and cellular function
- could be because of trauma, GI bleed
- Result of life-threatening conditions or trauma
- The body attempts to compensate for inadequate perfusion (homeostasis)
- **What are the consequences of inadequate blood flow to tissues?**

Pathophysiology of Shock- Cellular Level Effects

Do you
remember the
key differences
between
aerobic and
anaerobic
metabolism?

Metabolism changes to anaerobic due to hypoperfusion .

ATP product is affected. What by-product is also produced? How does this affect the cell?

Potassium in
sodium out

Na-K pump becomes impaired. What effect will this have? → swelling because H₂O follows Na

Cells are injured . What is released? → Inflammatory
Cell death occurs.

Pathophysiology of Shock- Compensation Mechanisms

- Goal: Maintain BP and CO

- **2 Key Mechanisms:**

1. Sympathetic nervous system

- What is the role of the SNS?
- What is released? → Epinephrine / Norepinephrine
- How will it help in shock?

2. Renin-angiotensin-aldosterone mechanism

- What is the effect of this mechanism?
- How will it help in shock?

You learned about
how the SNS and
RAA work in
anatomy class.
Now it is time to
apply it!

Compensatory Mechanisms in Shock

What are the other compensatory mechanisms?

SNS

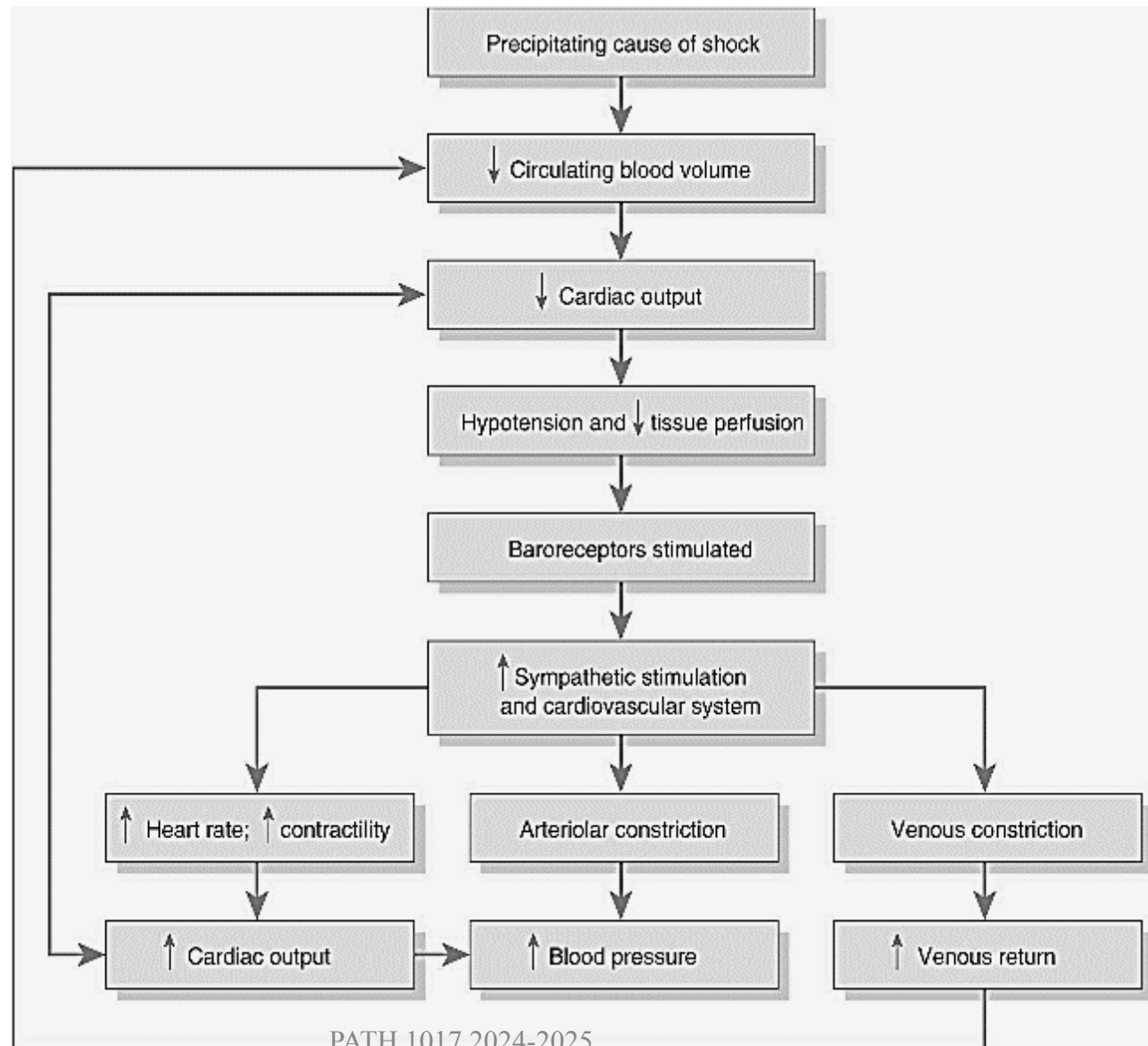
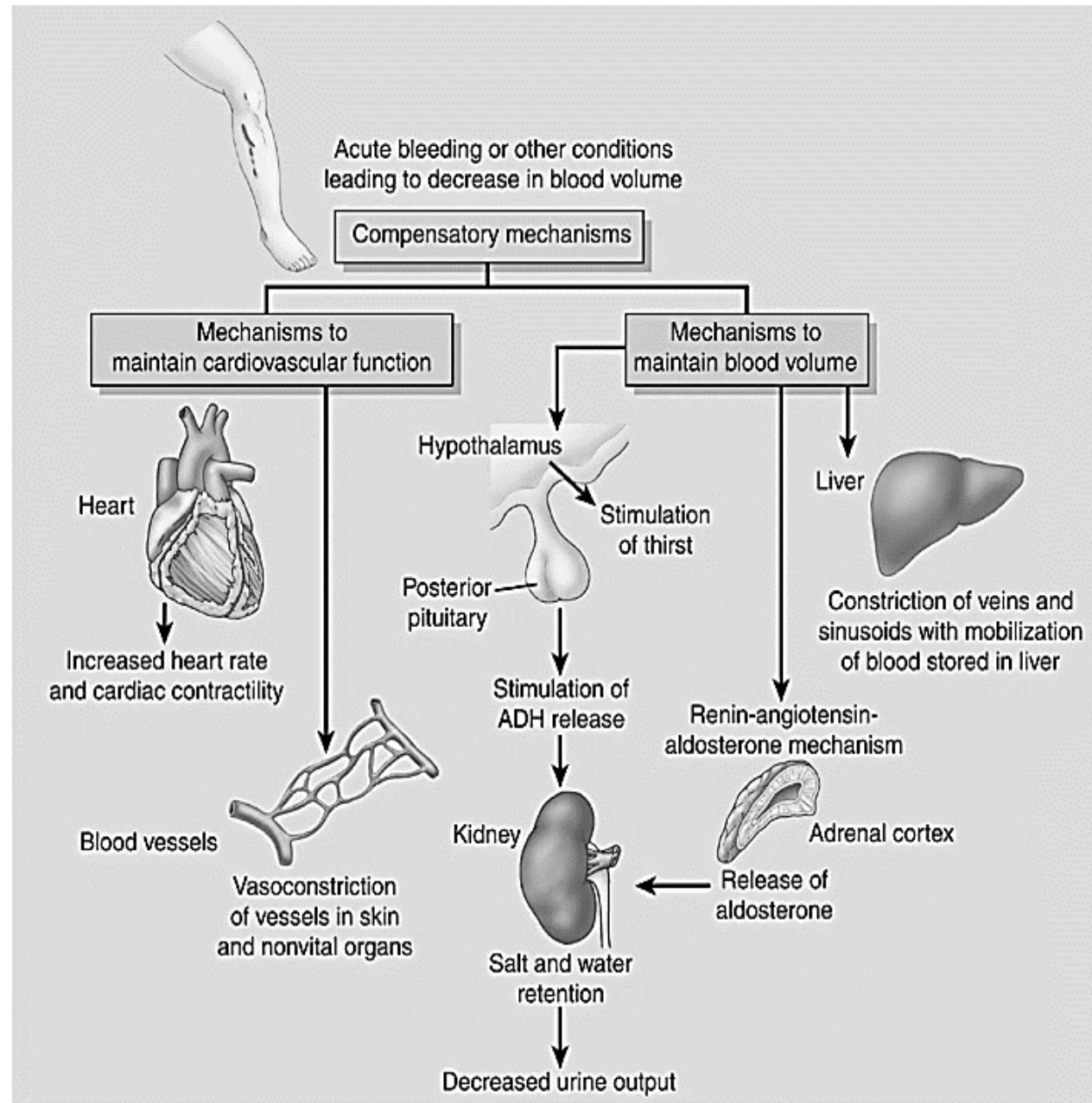


Fig 20-8- p503 Compensatory Mechanisms

Attempts to
Maintain
Circulatory
Function and
Blood Volume

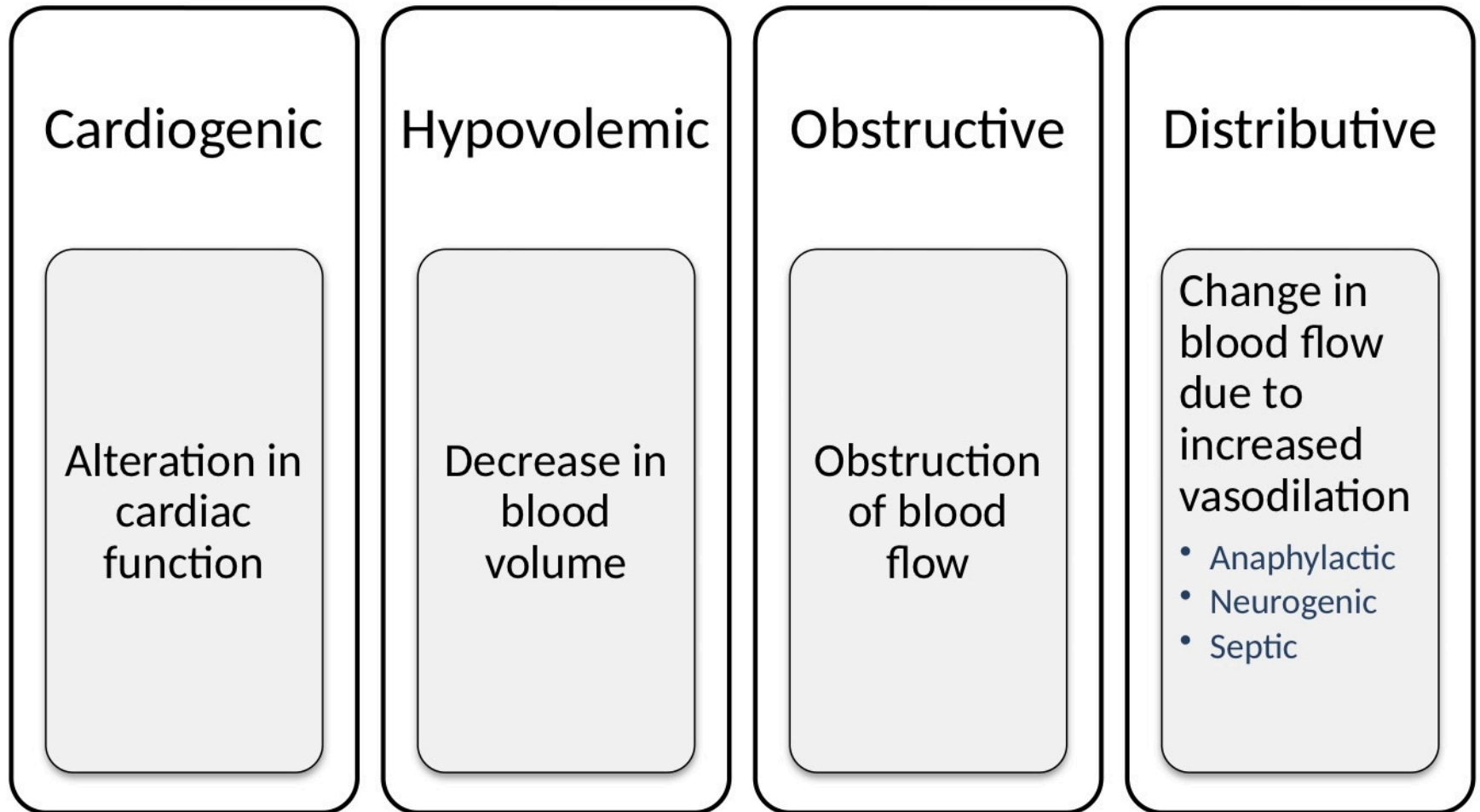
Think about the
SNS vs. RAA



Stages of Shock

	Compensatory	Progressive	Irreversible
Patho	Compensation mechanisms initiated in response to low perfusion	Compensation mechanisms can't keep up, hypoperfusion worsens, cell damage	Hypoperfusion severely damages multiple organs
Key Effects	• BP normal • HR ↑ • Cool clammy skin • Decreased bowel sounds • Urine output ↓ • Blood glucose ↑ • RR ↑ • Metabolic acidosis • Anxious or confused	• BP ↓ • HR ↑ ↑ • Cardiac ischemia • dysrhythmias • Urine output ↓ ↓ or absent • Hypoxia, pulmonary edema • Decreased LOC • Altered liver, GI, hematological etc.	• BP ↓ ↓ ↓ • Metabolic acidosis worsens • Multiple organs fail
Prognosis	Good with early intervention and correction of cause	Worsens- Depends on time to restore perfusion, damage etc.	Does not respond to treatment ... Death

Types of Shock (Overview)



Hypovolemic Shock

- Caused by diminished blood volume with decreased filling of the circulatory system

Caused by loss of:

- Whole blood
 - hemorrhage, internal bleeding
- Plasma
 - severe burns
- Extracellular fluid
 - GI losses from vomiting & diarrhea, third-spacing

Pathophysiology of Hypovolemic

How will the SNS compensate

- ↑ HR, Vasoconstriction

Decreased blood
volume



Decreased venous
return



Decreased stroke
volume



Decreased cardiac
output



Decreased tissue
perfusion

ck

What
compensatory
mechanisms
will be
initiated?

Identify the Manifestations of Hypovolemic Shock

- Early Signs:

- Late Signs:

Hypovolemic Shock Treatment

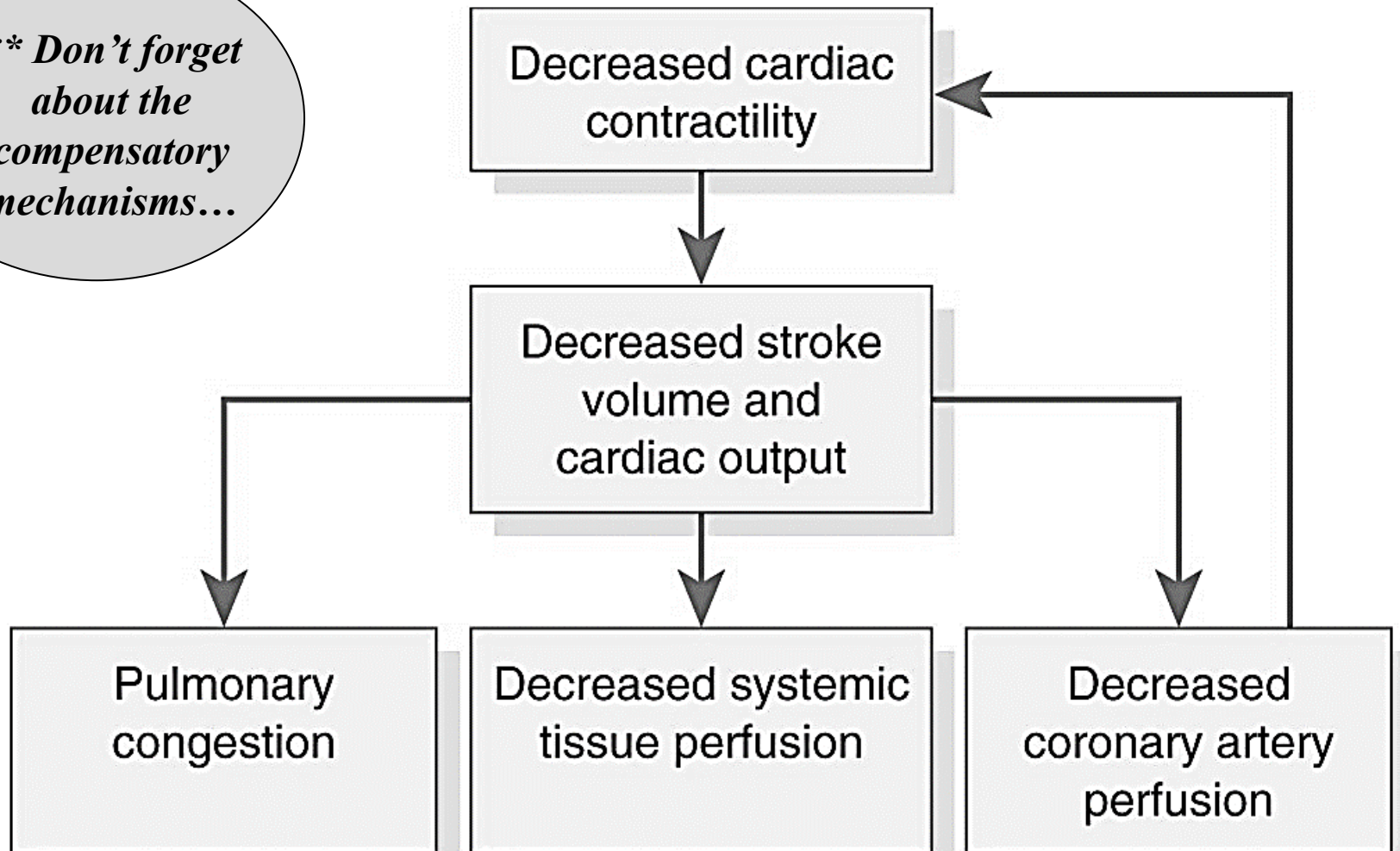
- Correct or control underlying problem
- Reestablish vascular volume
 - IV fluid, blood products
↳ to compensate blood loss
- Positioning
↳ Trendelenburg position
- Oxygen
- Medications (vasoactive)
↳ might be in exam
↳ vasoconstriction
 - may be needed to increase BP AFTER blood volume deficit has been corrected to help improve blood flow to kidneys and other organs
- Ongoing assessment & monitoring
 - Examples of key indicators of response?
↳ vital signs
 - Treatment complications?
↳ Foley catheter → assess urine function

Cardiogenic Shock

- Heart fails to pump blood adequately
- **Causes are coronary and non-coronary in origin:**
 - Damage to heart (MI) → most common
 - Ineffective pumping (arrhythmias)
 - Mechanical defects (i.e. septal damage, valve damage)
 - Post-cardiac surgery complications
 - End-stage cardiac conditions (i.e. CAD, cardiomyopathy)

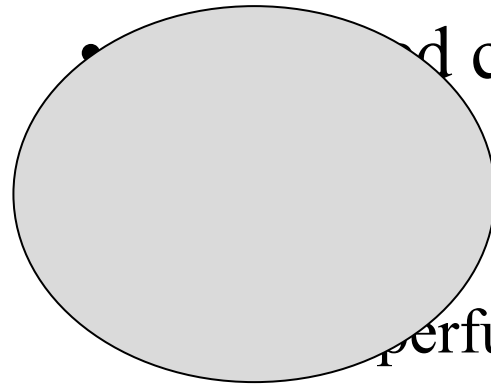
Pathophysiology of Cardiogenic Shock

***** Don't forget
about the
compensatory
mechanisms...***



Think about it...

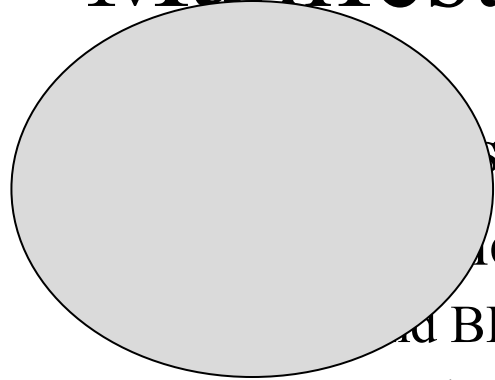
Cardiogenic



• Decreased cardiac output & lowers BP
• Effect on contractility? Preload? Afterload?
• Decreased cardiac output & lowers BP
– Effect on contractility? Preload? Afterload?
– Hypoperfusion despite normal volume

- Sympathetic system responds
 - Vasoconstriction increases resistance to blood flow.
- RAA is activated. Effects?
- Increased workload on heart worsens heart failure and affects coronary perfusion.

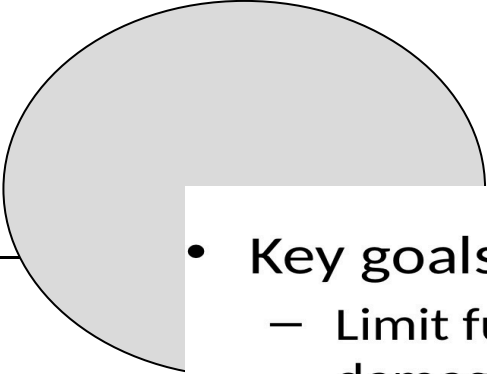
Manifestations of Cardiogenic Shock



- Hemodynamic instability
 - BP decrease
 - Dysrhythmias
 - Hemodynamic instability
 - MAP and BP decrease
 - Decreased urine output
 - Decreased urine output
- Similar appearances to heart failure
 - Cyanosis of lips, nail bed & skin
 - Pulmonary edema
- Changes in LOC

- Think about it...
What is the pathophysiology causing these manifestations?

Treatment of Cardiogenic Shock

- 
- Key goals:
 - Limit further myocardial damage and protect healthy myocardium

Improve CO

□ workload and O2 needs

□ coronary perfusion

Treatment of Cardiogenic Shock

- Oxygen and pain control
- Hemodynamic monitoring
- Medications (vasoactive)
- Fluid therapy
- Mechanical assistive devices (IABP)

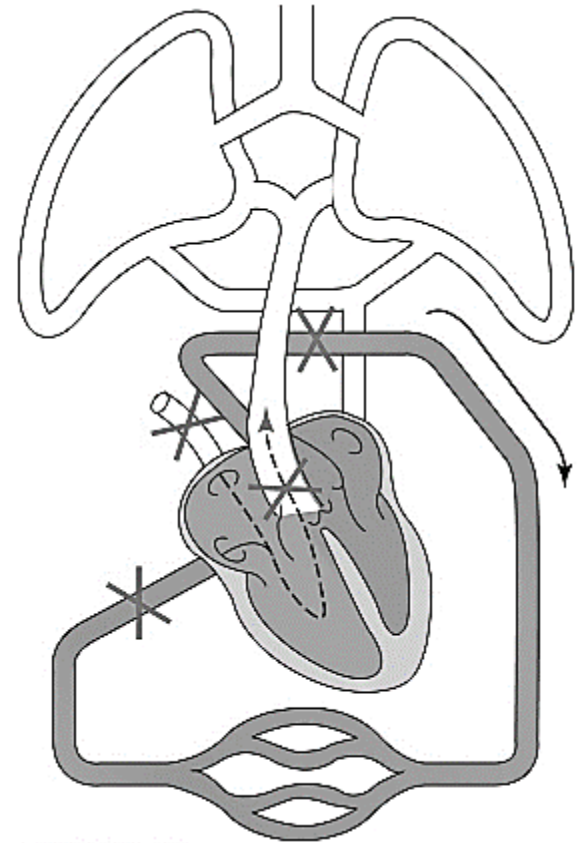
↳ Intra-aortic balloon pump

Don't forget to treat the cause of the cardiogenic shock...

Diastole → balloon inflates
Systole → balloon deflates

Obstructive Shock

- Due to mechanical impediment to flow of blood through central circulation (great veins, heart, & pulmonary circulation).
- Caused by:
 - Dissecting aortic aneurysm
 - Cardiac tamponade
 - Pneumothorax
 - Pulmonary embolism (most common)
- Creates increased pressure within the right side of heart
 - This leads to decreased blood return to heart



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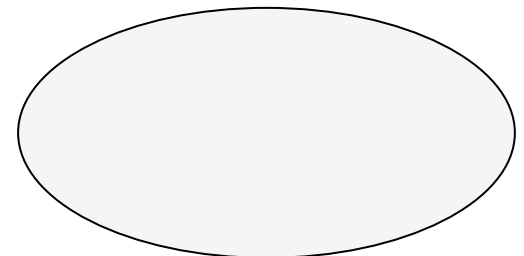
What will happen
to the CO?

Obstructive Shock

- Symptoms of right sided heart failure develop
- Treatment focus on reversing underlying cause
 - Examples?

* pulmonary embolectomy to remove clot from Pulmonary Embolism

* chest tube insertion to relieve tension pneumothorax



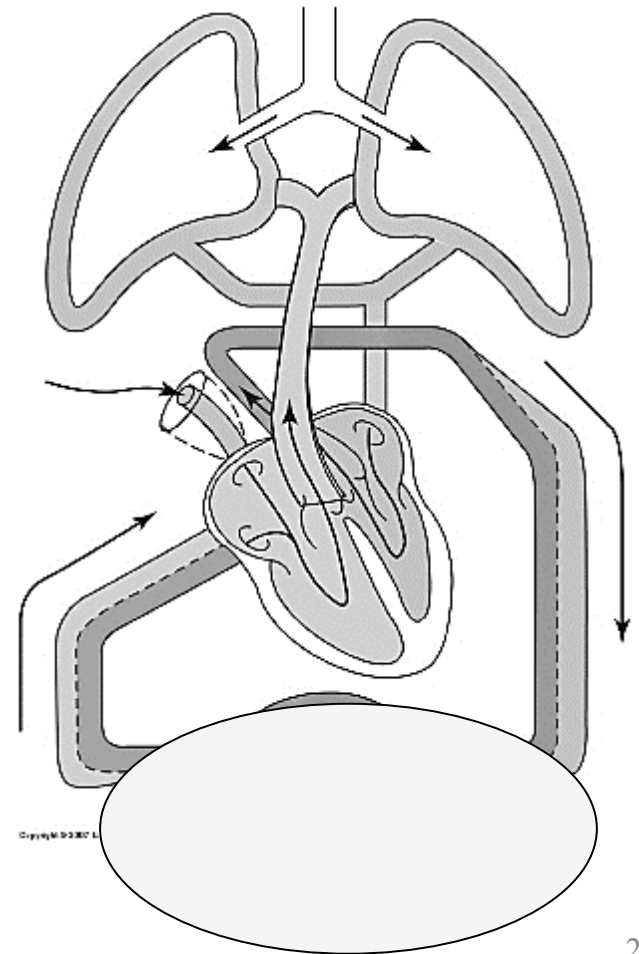
Distributive Shock

Vasodilation
shock
↓
Poor

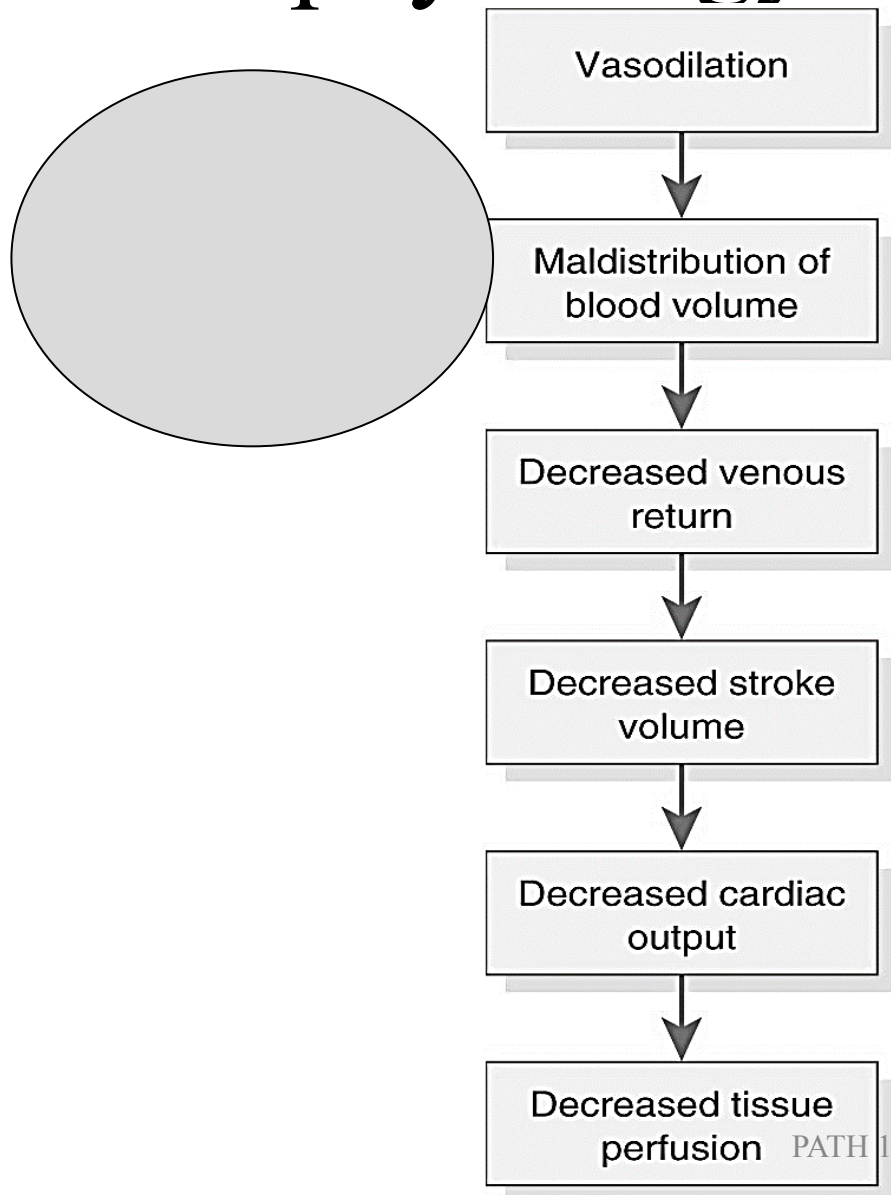
- Occurs when blood volume is abnormally displaced in the vasculature away from the heart and central circulation

Further classified into 3 types:

1. Neurogenic shock
2. Anaphylactic shock
3. Septic shock → common



Pathophysiology of Distributive Shock

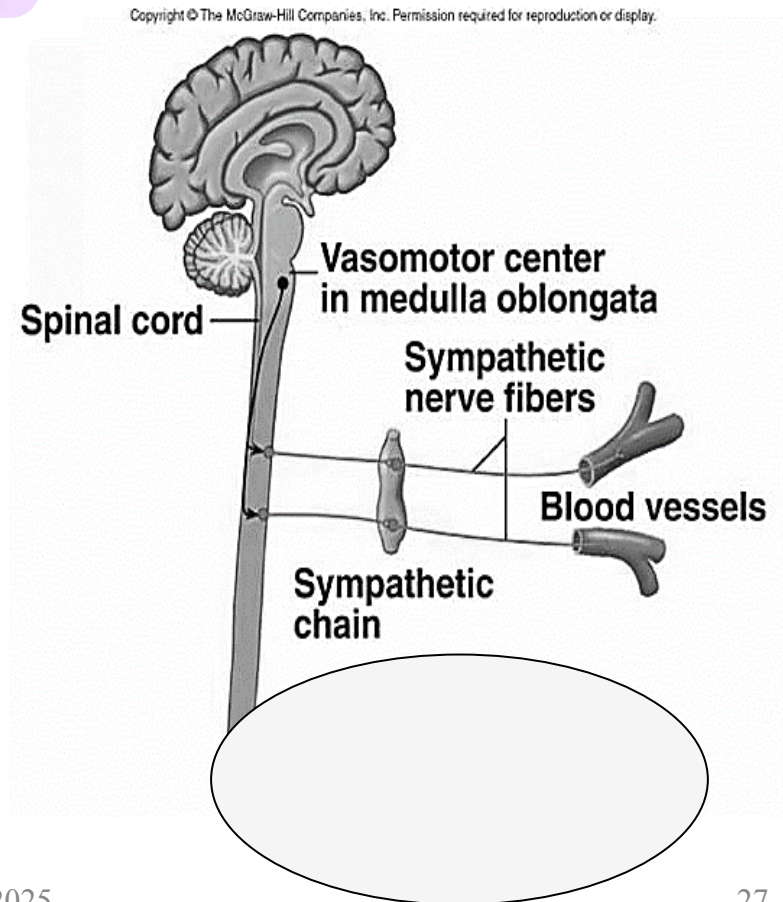


Compensatory mechanisms are either impaired or unable to overcome the effects of distributive shock

RAAS > NOT DOING THE RIGHT WORK
SNS

Neurogenic Shock

- Caused by vasodilation resulting from loss of sympathetic tone
- Can result from
 - spinal cord injury
 - spinal anesthesia
 - nervous system damage
 - lack of glucose



Neurogenic Shock

Interrupted SNS signals to blood vessels, baroreceptors and thermoregulation system

Relaxation of smooth muscle in blood vessel walls

Decreased systemic/peripheral vascular resistance (massive vasodilation)

Result- hypotension, decreased CO, bradycardia *in the 40s*
Leading to: **impaired peripheral perfusion**

Neurogenic Shock

why?

- Transient
 - Onset in minutes, duration up to 3 weeks
- Key Symptoms:
 - Bradycardia (baroreceptors cannot increase HR)
 - Hypotension (vasodilation, decreased CO) ↗ RAAS ↘ SNS > can't do anything cuz of the spinal injury
 - Warm flushed skin
 - Poikilothermy (loss of SNS thermoregulation ability)
- Key treatments:
 - Restore BP with: IV fluids, vasoactive
 - Restore HR -Medication to increase HR (atropine)

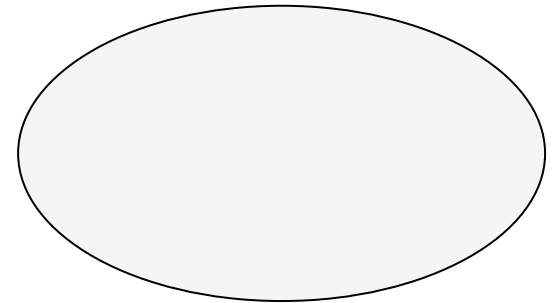
Anaphylactic Shock

- Occurs rapidly & is life-threatening
- Caused by severe allergic reaction
 - May occur with one exposure or after several exposures to an antigen
 - Involves an antigen-antibody reaction
 - Mast cells release Histamine which causes widespread vasodilation, increased capillary permeability
- Most common triggers :
 - Medication reactions (i.e.. penicillin)
 - Foods (nuts, shellfish)
 - Insect venom (bee stings)
 - Latex allergy

**What will happen
to the CO and
SVR?**

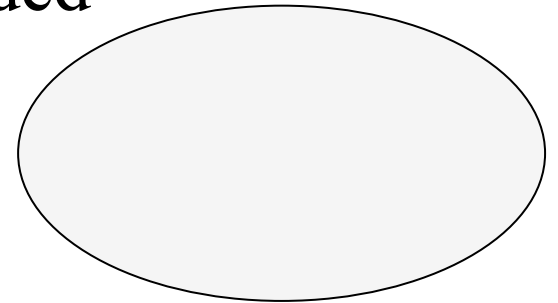
Anaphylactic Shock Manifestations:

- Anxiety
- Laryngeal edema (life-threatening)
- Bronchospasm
- Hypotension
- Tachycardia, weak pulse
- Pruritis (itching)
- Angioedema
- Urticaria (hives)
- Abdominal cramps (GI & uterine smooth muscle contraction)
- Loss of consciousness (late)



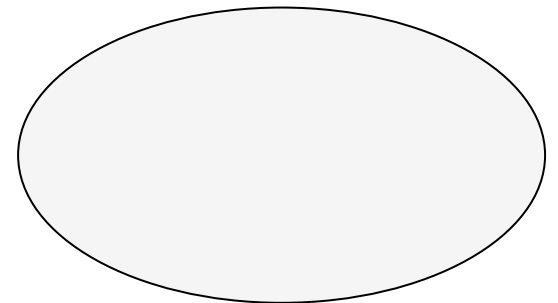
Anaphylactic Shock Treatment

- Remove triggering agent immediately
- ABCs
- Fluids – Epinephrine
- Drugs:
 - Epinephrine (i.e.. Epi-Pen)
 - Antihistamines (benadryl), corticosteroids
 - Vasoactive drugs – vasoconstriction
 - Bronchodilators
- Intubation & oxygen may be needed
- Prevention is best!



Septic Shock

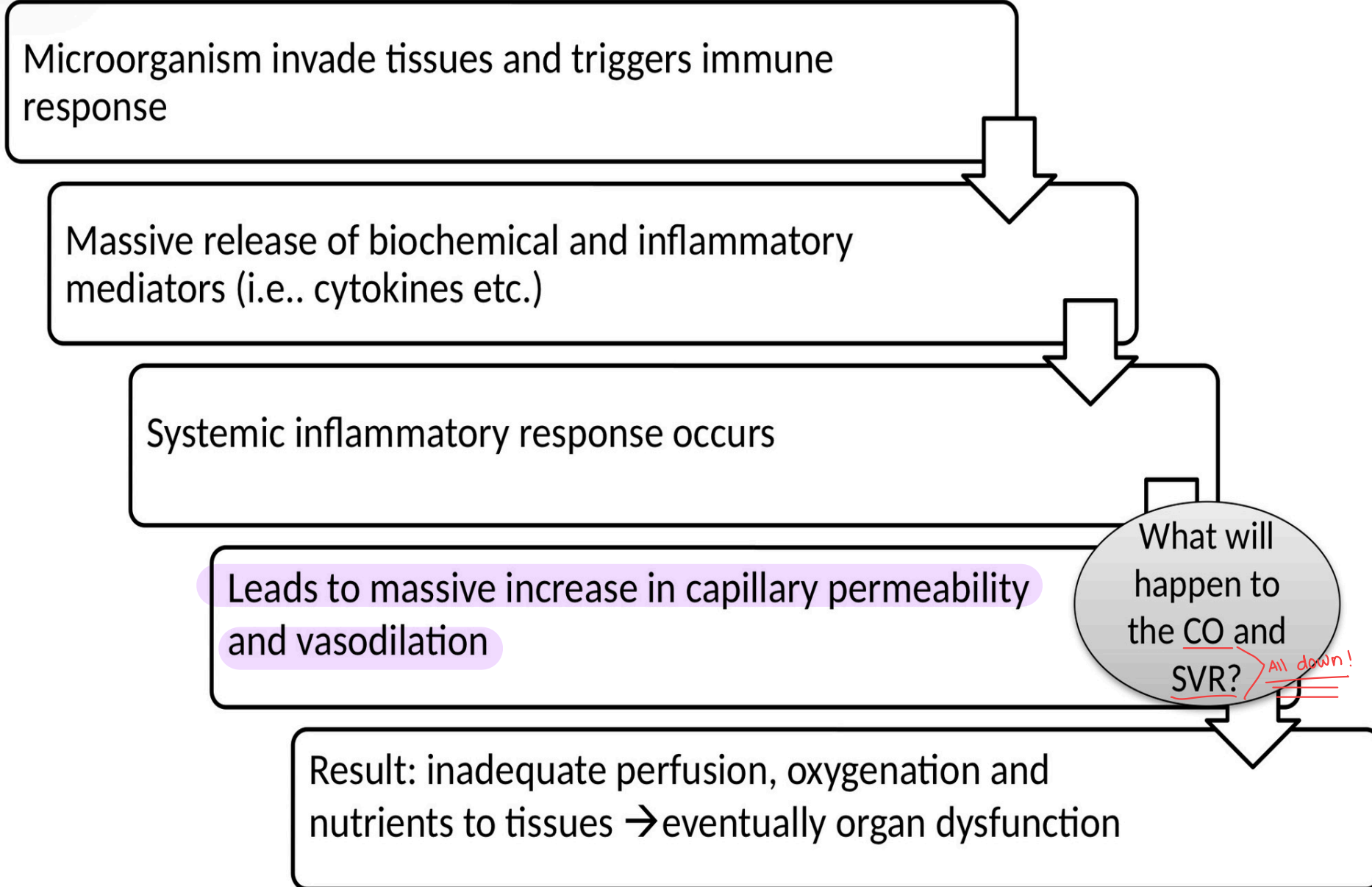
- Most common type of distributive shock
- Most commonly caused by gram-negative bacteria
 - Bacteremia and pneumonia common precursors
- Elderly patients at particular risk
- Prognosis: 28-50% die
- Prevention important



Septic Shock

* 3 concerns on vital signs

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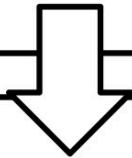
SEPTIC SHOCK - MANIFEST

Septic Shock- 2 Phases

Warm Phase

(hyperdynamic,
progressive)

- High CO
- Vasodilation ($\downarrow$ SVR)
- BP normal; HR $\uparrow$; RR $\uparrow$
- Fever, warm & flushed skin
- May have $\downarrow$ urine output; $\downarrow$ bowel sounds; confusion/agitation



Cold Phase

(hypodynamic,
irreversible)

- Low CO
- Vasoconstriction (cool, pale skin)
- $\downarrow$ temp; $\uparrow\uparrow$ HR; $\uparrow\uparrow$ RR
- Anuria
- Leads to organ failure, death

Septic Shock Treatment

- Identify cause (i.e.. C&S) and treat cause if possible
- Fluid replacement
- Oxygen (may require mechanical ventilation if shock is progressing)
- Pharmacological:
 - Antibiotics
 - Vasoactive drugs
 - Antipyretics
 - Corticosteroids → for inflammation
- Other aspects of treatment;
 - Nutritional support → IV
 - Control blood glucose levels (insulin tx)

Complications of Shock